

The Proved Impossibilities of D_IV⁵ — what one geometry forbids, and what it proves it cannot do

A theory's negatives are its spine: forcing forbids; fitting cannot.
Every item here is falsifiable, and any single violation kills the framework.

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The Proved Impossibilities of D_IV⁵

The one to lead with: a 0.0009% match we refuse

Before the list, the single sharpest item — because an honest instrument is proved honest by what it *turns away*. The charged-lepton Koide relation, $Q = (\Sigma m)/(\Sigma \sqrt{m})^2 = 2/3$, holds to $\approx 0.0009\%$ — the tightest coincidence in the neighbourhood. We do not claim it here, and our own instrument forbids the claim: the reading calculus is *structural* (it reads kinds — real/complex/quaternionic — not values), and Koide is a relation among mass *values*, provably outside what the instrument can read. We could have dressed it as a “census-and-rank” derivation; on the vet, that laundered a value through a structural instrument, and we struck it. A device that refuses its own best-looking number because its stated rules forbid the reading is an instrument, not an advocacy tool. (Koide is real; it may be derived by a *different* instrument — the mass/eigenvalue one — and sits on our ledger as identified-not-derived. Just not here.)

Why publish what a theory *cannot* do

A fitted model can absorb almost anything: a new particle, a new coupling, a missing length — you add a parameter and move on. A forced geometry cannot. Its structure *forbids* things, and those forbiddings are not hedges — they are predictions, sharper than most positive claims, because a single violation ends the theory. So we publish our impossibilities on purpose. They are the load-bearing test of whether BST is forcing or fitting: a theory that forbids nothing has explained nothing.

Three kinds of impossibility follow from the single geometry D_{IV}^5 ($= SO_0(5,2)/[SO(5)\times SO(2)]$): things that *do not exist*, structures the geometry *cannot supply*, and a scale it *cannot invent*.

I. Absences — things D_{IV}^5 forbids (the Five-Absence set, +1)

Each follows from the irreducibility of the one geometry: there is no larger structure for these objects to live in.

forbidden	mechanism	how it dies
No grand unification (GUT)	the three gauge factors are not on equal footing — two are internal symmetries the geometry derives, one has <i>no internal group at all</i> (Section II); nothing to unify	any high-scale unification (proton decay via X/Y, GUT-scale coupling meeting)
No proton decay	baryon number is a topological winding (a trefoil), a discrete conserved class, not an accidental symmetry	a single $\Delta B = 1$ event, at any scale
No supersymmetry	the geometry's fermion/boson content is fixed by the finite space; no room for superpartners	any superpartner
No magnetic monopoles	no relevant nontrivial π_2 of the substrate	a monopole
No dark-matter particle	the spectrum is the Standard-Model content, closed	a new stable particle in the halo
No sterile (right-handed) neutrino	ν_R has no orientable non-Möbius gauge coupling (no charge, no color), so its cycle cannot close on the non-orientable Shilov boundary (T1949)	a sterile ν at <i>any</i> mass (eV, keV, MeV, GeV–TeV)

A fitted Standard-Model extension can accommodate any one of these; BST cannot survive one. That asymmetry is the point.

II. Structural no-go — what the geometry cannot supply

These are theorems about the geometry’s own limits — the honest boundaries we do not paper over.

- No gluon generators — there is no internal color group (K1724). Decomposing the finite internal space under $K = SO(5) \times SO(2)$ is exact (Frobenius reality-type census): the three sectors land on the three real division algebras and contribute internal gauge generators $\{0, 1, 3\} \rightarrow 0+1+3 = 4 = \dim(U(1) \times SU(2))$. The color sector is a *real* 3-space; it contributes zero internal generators. So the geometry cannot geometrize color the way it geometrizes electroweak — $SU(3)$ is the standard gauge principle applied externally, not read off D_{IV}^5 . We prove the negative rather than hide it: the electroweak group is derived (its dimension is a *count*), and the color group is exactly what the geometry does *not* contain. (*The geometry still fixes the color number $N_c = 3$ — a geometric $SO(3)$ rotation of three real directions — and forbids v_R ; it just supplies no color gauge factor.*)
- The complete ruler theorem — the geometry fixes neither the *scale* nor the *shape* of the ruler (K1727/K1729). BST takes one dimensionful input (“the ruler”), and we can now prove that both halves of it are genuinely required, not shortfalls:
 - *No scale.* Zeta-regularizing the compactification (Kaluza–Klein) energy of the boundary as a function of its size *runs away* — no interior minimum, both branches — so the geometry cannot select a length. A dimensionless set cannot produce a quantity with units; exactly one dimensionful input (a Planck-scale unit) is forced, the same one GR pays as G and QM as $\hbar$. Proved.
 - *No forced shape.* The dimensionless boundary “shape” (the S^4/S^1 ratio) has no canonical value read off the geometry either: the candidate ratios either slide a mode-weight into a metric (retired), or trace back to a retired target-fit — there is no kernel-free intrinsic ratio the CR/conformal boundary selects. So the shape too is handed to physics, not forced. We bank the whole thing as a win: “you cannot get a length — or a boundary shape — from pure geometry” is proved. The single ruler is a theorem, not an apology, and BST is honest about *exactly* which one thing it takes.

III. Ontological no-go (structural)

- No CPT-mirror / no reversed-time copy. BST’s time is a single forward $SO(2)$ commitment flow; there is no geometric CPT-mirror sheet. The arrow of time is dynamical (the positivity of the K-Casimir spectrum), not a symmetry to be undone.

(Note — reclassified. “No fourth fermion generation” was previously listed here as an impossibility. It is not one: it is a positive reading of Invariant 3, the rank — a rank-2 domain has exactly $\text{rank} + 1 = 3$ support strata (Korányi–Wolf), so the geometry de-

rives three generations and no fourth. A fourth generation still falsifies BST, but as the violation of a derived count, not of a forbidding. Moved to the generating-invariants section, Part 2.)

Armed falsifiers — the current experimental front

The forbiddings above are not rhetorical; each has an experiment aimed at it. Two are decisive in this decade:

BST says	current limit	next experiment	BST's sharp claim
no baryon-number violation (no proton decay; no $n-\bar{n}$ oscillation)	$\tau(p \rightarrow e^+ \pi^0) > 2.4 \times 10^{34} \text{ yr}$ (Super-K); $n-\bar{n} \tau > 4.7 \times 10^8 \text{ s}$	Hyper- Kamiokande (~2027); ESS $n-\bar{n}$	null — baryon number is a topological winding; no $\Delta B=1$ (proton decay) <i>and</i> no $\Delta B=2$ ($n-\bar{n}$), at any scale

BST says	current limit	next experiment	BST's sharp claim
neutrino is Majorana (companion; forced)	$0\nu\beta\beta$ unseen; $m_{\beta\beta} < 28\text{--}122$ meV (KamLAND-Zen)	LEGEND-1000 (9–21 meV), nEXO (<15 meV)	two-sided. BST predicts $m_{\beta\beta} \in 1.5\text{--}3.7$ meV, <i>below</i> LEGEND-1000/nEXO reach → BST predicts null there (near-term null is <i>expected, not confirmation</i>). A detection by LEGEND-1000/nEXO would refute BST (a signal above ~4 meV contradicts $m_1 = 0$ / Normal Ordering). Confirmation needs a next-gen few-meV experiment reaching the 1.5–3.7 meV band (plus Project 8 / a KATRIN-successor measuring $m_1 = 0$ directly).
no sterile neutrino	no signal (LSND anomaly refuted by MicroBooNE)	SHiP, FCC-ee, reactor/oscillation	null at every mass
no fourth generation	consistent (Z-width, direct search)	any collider	null

Two of these are *sharp, opposite-signed* predictions on live experiments and cut both ways: a proton-decay or $n\text{--}\bar{n}$ event ends BST; and, on $0\nu\beta\beta$, a detection at LEGEND-1000/nEXO refutes BST while their null is expected — only a far-future few-meV experiment can *confirm* the Majorana prediction. We state the two-sided direction explicitly so a near-term null is not mis-read as a win, nor a near-term detection as a confirmation.

Part 2 — what these proofs let us *compute*: the impossibilities are instruments

An impossibility proof is not only a fence; it is a tool (Cal’s insight). Each “the geometry cannot do X” comes with a positive computational counterpart:

- The dimensional filter. “You cannot get a length from pure geometry” (the ruler theorem) is exactly the rule that tells you *which* quantities the geometry *can* fix: the dimensionless ones. Every BST derivation that lands (m_p/m_e , α , $\sin^2\theta_W$, the anomalies) is dimensionless; every quantity it honestly cannot fix carries the ruler. The no-scale proof is the *sorting instrument* that separates the two — it is why the ~ 150 dimensionless predictions are the right target and the masses-in-kilograms are not.
- The three generating invariants (the internal Standard Model, read from exactly three geometric data — counted once). The promotion test landed, and then *survived an attack that made it sharper*: the entire *discrete internal structure* of the Standard Model is generated by exactly three invariants of D_{IV}^5 .
 - Invariant 1 — the reality-type census of the finite space under $K = SO(5) \times SO(2)$: is each block real, complex, or quaternionic? It buys the gauge-group dimension ($\{0,1,3\} \rightarrow 4 = \dim(U(1) \times SU(2))$); no internal color group (the real block is a *spacetime* rotation, not a gauge symmetry — Coleman–Mandula); CP existence (the quaternionic spinor’s non-removable $\sqrt{-1}$ — *existence only*); anomaly-freedom (a self-conjugate rep has vanishing cubic anomaly — Elie); the custodial/Higgs structure.
 - Invariant 2 — the non-orientability of the Shilov boundary $(S^4 \times S^1)/\mathbb{Z}_2$: two-sided or one-sided? One bit. It buys parity violation and no sterile neutrino (ν_R cannot close on a Möbius edge).
 - Invariant 3 — the rank of $D_{IV}^5 (= 2)$: it buys the number of generations (rank + 1 = 3, and no fourth), the factor of 3 in the fermion count ($45 = 15 \times 3$), and the 3×3 structure of the mixing matrices.
 - Two facts need Invariants 1 and 2 together (attribution under test): *which* fermions are chiral, and that the neutrino is Majorana ($m_1 = 0$, $m_{\beta\beta} \in 1.5\text{--}3.7 \text{ meV}$; counted once with the no-sterile- ν absence).

This is structure, not spectrum, and it is a theorem (relative to the declared facts and operations — the honest form of “minimal”). The census can only see *kinds* (a fact is readable off it *iff* it is constant on the reality-type classes $\{\mathbb{R}, \mathbb{C}, \mathbb{H}\}$), so it is provably blind to the continuous *how-much* — masses, mixing angles, the CP magnitude live in a different instrument, and that blindness is a category boundary, not a gap. The whole discrete internal skeleton is under ~ 8 bits (census $\approx 4.75 +$ orientability 1 + rank), from three invariants, counted once — and the set was *attacked by our own referee and did not break* (the correction from two to three added the rank, which had been hiding in $45 = 15 \times 3$).

And the strongest form: across *all* such shapes it takes three independent

facts, but within our family (type IV) the three collapse to one — the rank is fixed at 2 by the family, and the census and orientability are both functions of the single dimension. So the honest headline is that the entire discrete Standard-Model skeleton is a function of one measured integer, $n_C = 5$ (the terminus theorem; the descent from three inputs to one is a *unification*, not a retreat).

- The absences as a closure certificate. “No GUT, no SUSY, no new particles” is the statement that the spectrum is *closed* — which is precisely what makes the content a finite, computable list (the 45 fermions, the 12 gauge bosons, the one Higgs) rather than an open-ended model. You can only compute anomalies, central charges, and a running coupling *because* the geometry forbids additions.

So the proved impossibilities are the flip side of the proved computations: the same fixed geometry that *forbids* is the one that *forces*. A theory that could add anything could compute nothing.

The meta-point

Every item above is (a) a *consequence of one fixed geometry*, not a tuned assumption; (b) *category-sorted* — a proved theorem (what the geometry cannot do) or a falsifiable prediction (what nature must therefore not contain), never blurred; and (c) *counted once* — the no-sterile-neutrino, the Majorana nature, and the $m_1=0$ all trace to one reality-type fact and are one line, not three. A single positive detection of any forbidden object — a proton decay, a fourth generation, a GUT signal, a sterile neutrino, a Dirac-only neutrino sector — ends BST. This is the opposite of a model that survives by absorption. We would rather be killed cleanly by one experiment than survive by adding parameters. The proved impossibilities are where that stance is cashed out — and, in Part 2, where the computations come from.

Outreach asset v0.1 (Lane C). Sources: Five-Absence Predictions Set (Casey-named); T1949 (ν_R Möbius exclusion) / T1984 (sterile- ν falsifier); K1724 (no internal color group / census-as-count); K1727 (no length from pure geometry / Casimir runaway); T2525 (three generations = rank+1); BST positive-time ontology. Keeper audits; ships on Keeper-PASS + Cal-vet + both voices + Casey GO. Nothing pushed; CP existence-only.

— Lyra, Thursday 2026-08-20 (date-verified).